AgriCopilot: An Explainable, Multimodal Precision-Agriculture Decision Support System Integrating Deep Learning, Dempster–Shafer Evidence Fusion and Retrieval-Augmented Generation

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***Abstract*—Smallholder farmers have access to more data than ever—weather feeds, soil maps, Sentinel-2 imagery, mandi prices—but these streams sit in separate silos, and the machine-learning tools built on them usually return unexplained point predictions. This paper presents AgriCopilot, a full-stack, explainable decision support system that fuses four data modalities through six model families. Two EfficientNetV2-S classifiers diagnose 55 crop-disease classes (94.82% on Indian field images, 99.49% on PlantVillage); a soft-voting ensemble recommends one of 22 crops (99.55%); a super-ensemble grades soil fertility (95.84% five-fold CV); and an XGBoost regressor predicts yield ($R^2 = 0.9632$ on a synthetic benchmark). Every output carries an explanation: Grad-CAM for images, TreeSHAP for tabular models, split-conformal intervals for yield, and Dempster–Shafer (DS) belief–plausibility intervals for six agricultural risks, with explicit "ignorance" mass for missing sensors. A retrieval-augmented generation (RAG) copilot grounded in TNAU/ICAR material gives cited advice. In a four-configuration ablation, adding weather, soil and NDVI to vision cuts risk-index error by 64.5% and raises risk-category agreement from 64.2% to 92.4%. Results are compared with recent (2023–2026) literature, and the limitations of the evaluation are stated openly.**

***Index Terms*—Precision agriculture, explainable AI, EfficientNetV2, Dempster–Shafer theory, TreeSHAP, Grad-CAM, conformal prediction, retrieval-augmented generation, multimodal fusion, SDG.**

I. Problem Identification

Title: *Fragmented Agricultural Intelligence: The Absence of Unified, Explainable, Multimodal Decision Support for Smallholder Farmers in Data-Rich Agrarian Ecosystems.*

Agriculture employs over 40% of India's workforce and contributes roughly 18% of GDP [1]. Open data are abundant: Sentinel-2 satellites, Open-Meteo and NASA POWER climate APIs, ISRIC SoilGrids and Agmarknet prices. Yet a farmer deciding what to sow, whether a leaf spot needs spraying, or whether to irrigate must still combine these sources by hand. Three problems follow.

1) **Data fragmentation.** Weather, soil, satellite, market and crop-health data differ in format, time step and spatial resolution. We know of no open platform that aggregates all nine streams used here in one pipeline.
2) **Black-box opacity.** Most disease and yield models give a label or a number but no reason. For decisions on pesticide, fertiliser or crop choice, an unexplained output is hard for a farmer or extension officer to trust or audit.
3) **Missing uncertainty.** Point predictions hide how much the system does not know. When an API is offline or a satellite tile is cloudy, a conventional pipeline silently imputes values and stays confident.

II. Objectives and Sub-Objectives

A. Primary Objective

To design, implement and evaluate an end-to-end explainable decision support system that combines multimodal data fusion, multi-task machine learning, formal uncertainty quantification and conversational advisory for smallholder farmers.

B. Sub-Objectives

1) **Multi-source ingestion:** build an asynchronous pipeline over nine APIs (Open-Meteo, NASA POWER, SoilGrids, Agromonitoring NDVI, Agmarknet, TNAU cost baselines, ICAR crop calendars, Tamil Nadu soil maps, Tavily search) with explicit handling of missing data.
2) **Disease diagnosis:** train two EfficientNetV2-S models (17 Indian field classes; 38 PlantVillage classes) with confidence-based routing.
3) **Tabular prediction:** build ensembles for 22-crop recommendation and 3-class soil fertility.
4) **Yield with uncertainty:** benchmark XGBoost, random forest and OLS on 21 covariates; add TreeSHAP and 90% split-conformal intervals.
5) **Risk fusion:** implement DS evidence theory over six risk dimensions with ignorance mass for missing inputs.

TABLE I
SDG ALIGNMENT OF AGRICOPILOT

SDG	Target	Contribution
2 Zero Hunger	2.3: raise productivity and income of small-scale producers	Crop recommendation, yield prediction with intervals and early disease detection support higher yield and lower loss per hectare.
13 Climate Action	13.1: resilience to climate-related hazards	DS fusion of drought, flood and pest risk with explicit uncertainty; weather and NDVI monitoring give early warning.
15 Life on Land	15.3: land-degradation neutrality	Soil-fertility grading, targeted fertiliser advice and NDVI monitoring discourage nutrient mining.
12 Responsible Consumption	12.4: sound chemical management	Precision fertiliser and spray advice reduces chemical over-use.
9 Industry, Innovation	9.5: enhance technological capability	Open, reproducible architecture combining evidence theory with deep learning.

6) **RAG copilot:** build a ChromaDB-backed assistant over TNAU/ICAR text with an LLM back end and live web search.
7) **Visual explainability:** overlay Grad-CAM heat maps to show that the CNN attends to lesions rather than background.

III. PROBLEM STATEMENT

Smallholder farmers in tropical agrarian economies need one accessible web platform that (i) ingests live data from at least nine heterogeneous agricultural sources; (ii) performs four inference tasks—disease classification, crop recommendation, soil-fertility grading and yield prediction; (iii) explains each output with a method suited to its data type; (iv) quantifies predictive uncertainty with statistical guarantees; and (v) answers questions in natural language, grounded in authoritative sources.

Explanation. Existing work covers these needs piecemeal. Deep CNNs reach very high accuracy on leaf images but rarely explain themselves or use weather and soil context [2], [3]. Crop-recommendation papers have recently added SHAP/LIME [11], [12] but still treat the problem as a seven-feature table. Yield studies report point errors [16], [17]. Agricultural RAG systems are only now appearing [26]. Surveying the papers reviewed in Section V, each addresses at most two of the five requirements. The gap is therefore integration: a system in which the vision, tabular, risk and language components share one context, and in which the system knows, and says, when it lacks data.

IV. RELEVANT SUSTAINABLE DEVELOPMENT GOALS

Table I maps the system to the United Nations SDGs.

V. LITERATURE SURVEY

A. Crop-Disease Detection

Mohanty *et al.* [2] reported 99.35% accuracy with GoogLeNet on 38 PlantVillage classes, and Ferentinos [3] reached 99.53% over 58 classes. Both used laboratory-style images and offered no explanation. Recent work has added XAI and exposed the lab-to-field gap. Pal *et al.* [4] proposed Mob-Res (3.51 M parameters, MobileNetV2 with residual blocks) with Grad-CAM, Grad-CAM++ and LIME; it scored 99.47% on PlantVillage but only 88.00% on a sugarcane field set. Alhammad *et al.* [5] used a VGG16 with Grad-CAM on potato leaves (98%, three classes). Fandi *et al.* [6] paired EfficientNetV2-S with ResNet50 and reached 97.16% on a binary healthy-versus-diseased task, against 89.76% for EfficientNetV2-S alone in their setup. Barbedo [7] had earlier argued that lesion-level analysis and interpretability are needed for field use.

B. Crop Recommendation

Early systems used random forests or ensembles on the 22-crop soil–weather table [8]–[10]. Mahmud *et al.* [11] tuned a random forest with a genetic algorithm (99.3%) and explained it with LIME and SHAP. Bouni *et al.* [12] paired random forests with SHAP/LIME for interpretable recommendation. Kumar *et al.* [13] stacked XGBoost, LightGBM, CatBoost and TabNet with Monte-Carlo-dropout and NGBoost uncertainty (99.39%), though their own table shows a plain random forest slightly higher (99.70%). The lesson is that this benchmark is saturated, and the contribution lies in explanation and uncertainty rather than the last tenth of a percent.

C. Yield Prediction

Khaki and Wang [14] used deep networks for county-level maize and soybean yield, and Shahhosseini *et al.* [15] showed that stacked ML ensembles improve corn-yield forecasts. Li *et al.* [16] showed that how the yield trend is removed matters as much as the model: global de-trending lifted XGBoost R^2 by about 20% over no trend handling. Jagan Mohan *et al.* [17] applied SHAP and LIME to rice yield but reported no uncertainty estimate. Distribution-free conformal methods are established in agricultural vision [18] but, to our knowledge, rarely paired with SHAP for yield.

D. Explainability and Evidence Theory

Grad-CAM [19] and SHAP [20] are the standard visual and tabular explainers. Dempster–Shafer theory [21], [22] represents ignorance separately from disbelief; it is well known in remote-sensing fusion [23], though its normalisation can behave counter-intuitively under high conflict [24]. We found no agricultural platform that applies DS fusion with explicit sensor-dropout mass across several risk dimensions.

E. Retrieval-Augmented Generation

RAG [25] grounds an LLM in retrieved passages. In agriculture, AgriRegion [26] re-ranks extension documents by geographic proximity and a fine-tuned LLaMA 3 generator. It improves regional relevance, but it neither connects to live sensor context nor explains its predictions.

F. Summary Table

Table II summarises the studies above. ✓ = provided, ✗ = not reported.

TABLE II
LITERATURE SUMMARY AND POSITIONING OF THE PRESENT WORK

Ref.	Authors (Year)	Method	Task / Data	Reported result	XAI	Multimodal	Uncertainty	RAG
[2]	Mohanty (2016)	GoogLeNet	38-class disease, PlantVillage	99.35% acc.	✗	✗	✗	✗
[3]	Ferentinos (2018)	VGG-type CNN	58-class disease	99.53% acc.	✗	✗	✗	✗
[4]	Pal (2025)	Mob-Res (MobileNetV2+res.)	PlantVillage / field sugarcane	99.47% / 88.00%	Grad-CAM, LIME	✗	✗	✗
[5]	Alhammad (2025)	VGG16	3-class potato leaf	98% acc.	Grad-CAM	✗	✗	✗
[6]	Fandi (2026)	EffNetV2S+ResNet50	Binary leaf health	97.16% acc.	✗	✗	✗	✗
[11]	Mahmud (2024)	RF + genetic algorithm	22-crop recommendation	99.3% acc.	SHAP, LIME	✗	✗	✗
[12]	Bouni (2024)	RF	Crop recommendation	not stated in abstract	SHAP, LIME	✗	✗	✗
[13]	Kumar (2026)	XGB/LGBM/Cat/TabNet stack	22-crop recommendation	99.39% (RF 99.70%)	SHAP	✗	MC-dropout, NGBoost	✗
[16]	Li (2024)	XGBoost + de-trending	US county maize/soy yield	R^2 +20% vs. baseline	✗	Partial	✗	✗
[17]	Jagan Mohan (2025)	DT/RF/LightGBM	Rice yield	inconsistent in paper	SHAP, LIME	✗	✗	✗
[26]	Fanuel (2025)	Region-aware RAG	Agricultural QA	RAGA-prec. 0.86	Citations	✗	✗	✓
Ours	**AgriCopilot (2026)**	**EffNetV2 + XGB + DS + RAG**	**Multi-task, 9 sources**	**94.8–99.6%; R^2 0.963**	**Grad-CAM, SHAP, DS**	✓	**Conformal + DS**	✓

VI. NOVELTY

1) **One platform, nine sources, four modalities.** Weather, soil, satellite, market, climatology, costs, calendars, regional soil maps and web search feed one inference pipeline, and the outputs are fused by evidence theory rather than ad-hoc weighted averaging.
2) **Three-level explainability.** Grad-CAM answers *where* on the leaf; TreeSHAP answers *which variables* drove a number; DS belief–plausibility intervals answer *how sure* the system is given the data it actually had. Each of the recent XAI studies in Table II uses one or two of these, not all three.
3) **DS risk fusion with sensor-dropout modelling.** A missing API or a cloud-covered tile moves mass to $m(\Theta)$ instead of being imputed. The interface can then distinguish "the field is safe" from "there is not enough data to say".
4) **Confidence-based dual-model disease routing.** A lab-trained 38-class model and a field-trained 17-class model are chained so the farmer need not name the crop, covering 55 classes.
5) **Conformal yield intervals with SHAP.** A 90% distribution-free interval is shown beside the exact additive attribution of the point estimate.
6) **Grounded copilot with live context.** A RAG assistant over curated TNAU/ICAR text plus web search receives the current field state (location, soil, weather) as context.

VII. METHODOLOGY

A. Architecture

AgriCopilot has three tiers (Fig. 1): a Next.js presentation layer, a FastAPI back end exposing more than 15 REST endpoints, and an intelligence engine of PyTorch, scikit-learn and XGBoost models, a ChromaDB vector store and an LLM (Groq cloud, or local Ollama/Mistral as fallback).

TABLE III
DATA SOURCES

#	Source	Features
1	Open-Meteo	temperature, humidity, wind, rain, UV, forecast
2	ISRIC SoilGrids 2.0	clay, sand, organic carbon, CEC, bulk density, N–P–K, pH
3	Agromonitoring	Sentinel-2 NDVI, cloud coverage
4	NASA POWER	solar radiation, growing degree days, dew point
5	Agmarknet (data.gov.in)	mandi commodity prices
6	TNAU Agritech	per-hectare cultivation cost
7	ICAR / FAO	kharif/rabi sowing and harvest windows
8	TN Mannvalam/NBSSLUP	district soil-type map
9	Tavily	live agricultural extension search

B. Data Ingestion

Table III lists the nine sources. Each call has a timeout and a fallback; when a source fails the pipeline returns `None` and the DS engine assigns that source's mass to ignorance, rather than substituting a default.

C. Model 1: Dual Disease Classifier

Both classifiers use an EfficientNetV2-S backbone [27]. **Model 1A** (17 classes: corn, potato, rice, sugarcane, wheat) uses a `Dropout(0.2)+Linear(1280,17)` head and was trained on 13,324 field photographs with variable light, occlusion and soil background. **Model 1B** (38 classes, 14 crops) uses an MLP head (1280→512→128→38 with BatchNorm, SiLU, dropout 0.3) and was trained on 54,303 laboratory scans.

Routing: 1B runs first; if it recognises a horticultural leaf with confidence ≥ 0.80 its diagnosis is returned, otherwise 1A processes the image. *Training:* phase 1 freezes the backbone (AdamW, lr 10^{-3}, cross-entropy with label smoothing 0.1); phase 2 unfreezes the top four MBConv stages with cosine-annealed learning rate, RandAugment and CutMix. *Explanation:* Grad-CAM [19] is taken from the last convolutional block.

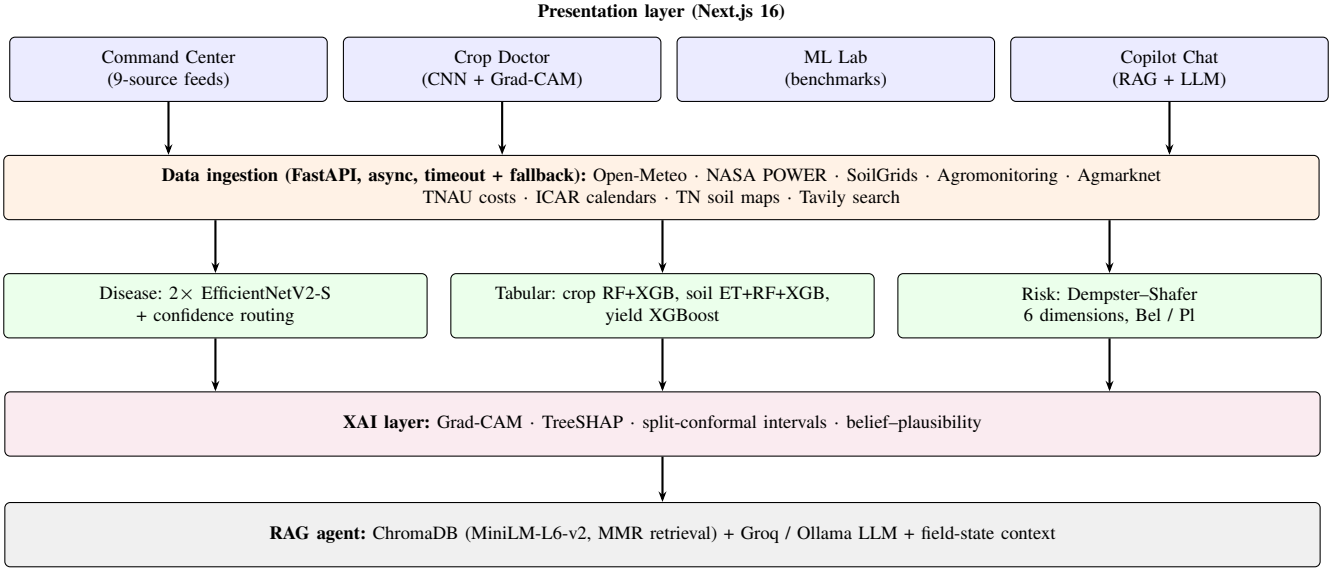


Fig. 1. System architecture of AgriCopilot.

D. Models 2 and 3: Crop Recommendation and Soil Fertility

Crop recommendation is a soft-voting ensemble of a random forest (200 trees, depth 16) and XGBoost [28] (150 rounds, lr 0.08) over seven inputs (N, P, K, temperature, humidity, pH, rainfall) for 22 crops; the data are 2,200 samples with a stratified 80/20 split. **Soil fertility** is a weighted soft vote of ExtraTrees (500, 0.4), random forest (500, 0.3) and XGBoost (350, 0.3) over 12 physico-chemical parameters plus 15 engineered features (nutrient ratios, interactions, threshold scores), on 880 samples and three classes (low, optimal, high). A rule table maps each class to fertiliser advice.

E. Model 4: Yield Prediction with Uncertainty

XGBoost, random forest and OLS are benchmarked on 21 covariates (weather 8, soil 9, crop biology 3, NDVI 1). TreeSHAP gives exact Shapley values

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!}\,[f(S \cup \{i\}) - f(S)], \tag{1}$$

which satisfy $f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i(x)$. Split-conformal prediction [29] uses a held-out calibration set to compute the $(1-\alpha)$ quantile $q_{1-\alpha}$ of absolute residuals and returns

$$\hat{C}_{1-\alpha}(x) = [\hat{y}(x) - q_{1-\alpha},\ \hat{y}(x) + q_{1-\alpha}], \quad \alpha = 0.10, \tag{2}$$

which covers the true value with probability $\geq 90\%$ for exchangeable data.

F. Dempster–Shafer Risk Fusion

For each risk the frame of discernment is $\Omega = \{\text{Risk}, \text{Safe}\}$. Each data source assigns a basic probability assignment with

$$m(\{R\}) + m(\{S\}) + m(\Theta) = 1, \quad m(\emptyset) = 0, \tag{3}$$

where $m(\Theta)$ is ignorance (missing data, cloud, API downtime). Two independent sources combine by Dempster's rule [21]:

$$m_{1,2}(A) = \frac{\sum_{B \cap C = A} m_1(B)\, m_2(C)}{1 - K}, \tag{4}$$

$$K = \sum_{B \cap C = \emptyset} m_1(B)\, m_2(C). \tag{5}$$

The output is an interval, with $\text{Bel}(R) = m(\{R\})$ and $\text{Pl}(R) = m(\{R\}) + m(\Theta)$, so that $\text{Pl} - \text{Bel} = m(\Theta)$ measures data ambiguity directly. Six risks are evaluated (Table IV). Because $K \to 1$ makes Dempster's normalisation unstable [24], the engine reports K and flags high-conflict cases.

TABLE IV
RISK DIMENSIONS AND EVIDENCE RULES

Risk	Sources	Evidence rule
Drought	weather, NASA, NDVI	$\text{ET}_0 - \text{rain} > 30$ mm; NDVI < 0.25
Flood	7-day rain, soil texture	rain > 100 mm and clay $> 45\%$
Pest/disease	humidity, temp., vision	RH $> 85\%$ at 22–30 °C plus classifier output
Soil degradation	SoilGrids, TN maps	pH < 5.0 or > 8.5; OC < 5 g/kg
Market price	Agmarknet, TNAU	negative gross-margin ratio
Yield loss	conformal yield, history	prediction vs. 5-year district benchmark

G. RAG Copilot

Eleven curated collections from the TNAU Agritech portal, ICAR handbooks and IPM spray schedules (ten crops: rice, maize, millets, sugarcane, banana, cotton, groundnut, tomato, onion, mango) are embedded with `all-MiniLM-L6-v2` into a persistent ChromaDB store using cosine similarity. Retrieval is top-k with Maximal Marginal Relevance for diversity; Tavily search adds live information. The LLM is gpt-oss-20b on Groq, with local Mistral through Ollama as a switchable fallback.

TABLE V
BACK-END ROUTER MODULES

Module	Endpoints	Function
Core	`/api/health`, `/api/analyze`, `/api/chat`	health, field analysis, copilot
ML	`/api/ml/yield/predict`, `/crop/recommend`, `/soil/fertility`	yield+SHAP+CI, crop, soil
Vision	`/api/diagnose-disease`	dual CNN + Grad-CAM
Risk	`/api/risk-dashboard`	DS six-risk fusion

TABLE VI
DATASETS USED

Dataset	Samples	Classes/features	Model
Indian crop diseases	13,324 img	17 classes	1A
PlantVillage	≈54,300 img	38 classes	1B
Crop recommendation	2,200	7 feat., 22 crops	2
Soil fertility	880	12 (+15 eng.)	3
Yield (**synthetic**)	2,000	21 covariates	4

VIII. IMPLEMENTATION AND DESIGN

A. Technology Stack

The front end is Next.js 16 (App Router) with hand-written CSS, about 2,500 lines in 7 files. The back end is FastAPI on Uvicorn, over 15,000 lines in 30+ files, serving asynchronous REST. Models run in PyTorch, scikit-learn and XGBoost, with eight or more checkpoints. ChromaDB persists the vectors, and the LLM is selected by an environment flag.

B. Front-End Design

Four pages share a React context (`FieldContext`) holding latitude, longitude, crop, soil type and field size, and a Leaflet map picker that also resolves the Tamil Nadu soil type.

- **Command Center** (three tabs: Overview, Risk, ML Lab): live weather, soil and market cards; NDVI gauges; crop probabilities; yield with SHAP waterfall; soil fertility with fertiliser advice.
- **Crop Doctor:** image upload or camera capture, diagnosis, Grad-CAM overlay, confidence dial and LLM treatment advice.
- **Model Performance Lab:** five tabs of accuracy, loss curves, confusion matrices and benchmarks.
- **Copilot Chat:** multi-turn streaming chat with citations and field context.

C. Back-End API

Four routers expose the services (Table V).

D. Datasets and Checkpoints

Table VI lists the datasets. The two disease checkpoints are 229 MB (1A) and 84 MB (1B); the crop, soil and yield models total under 60 MB.

TABLE VII
DISEASE CLASSIFICATION: OURS VS. LITERATURE

Method	Data	Acc. (%)	XAI
Mohanty'16 [2]	PV, 38 cl.	99.35	✗
Ferentinos'18 [3]	own, 58 cl.	99.53	✗
Pal'25 Mob-Res [4]	PV, 38 cl.	99.47	✓
Pal'25 Mob-Res [4]	field sugarcane	88.00	✓
Alhammad'25 VGG16 [5]	potato, 3 cl.	98.0	✓
Fandi'26 hybrid [6]	PV, binary	97.16	✗
Ours 1B	PV, 38 cl.	**99.49**	✓
Ours 1A	Indian field, 17 cl.	**94.82**	✓

Macro-F1: 1A 0.945, 1B 0.995. Test loss: 0.1842 / 0.0210. PV = PlantVillage.

TABLE VIII
CROP RECOMMENDATION: OURS VS. RECENT WORK (22 CROPS)

Method	Acc. (%)	XAI	Uncert.
Mahmud'24 RF+GA [11]	99.3	✓	✗
Kumar'26 stacked ensemble [13]	99.39	✓	✓
Kumar'26 random forest [13]	99.70	✗	✗
Kumar'26 XGBoost [13]	99.09	✗	✗
Ours RF+XGB soft vote	**99.55**	✓	✓(DS)

IX. RESULTS AND COMPARISON WITH RECENT WORK

All figures below are from our own held-out evaluations. Where we compare with published papers, protocols, datasets and class counts differ, so the comparison shows where AgriCopilot stands, not a controlled ranking.

A. Disease Classification

Table VII gives our results and a comparison with recent papers. Model 1B (99.49%) matches the best lab-image results in the literature, and Model 1A reaches 94.82% on 17 classes of real field photographs. The 4.7-point gap between the two agrees with the 99.47% → 88.00% lab-to-field drop reported by Pal *et al.* [4], and confirms that field data are harder. Unlike the earlier CNNs [2], [3], both models are explained by Grad-CAM.

B. Crop Recommendation

On 440 test samples the soft-voting ensemble made 2 errors (rice→jute and a lentil miss): accuracy 99.55%, weighted precision 99.57%, recall 99.55%, F1 99.55%. Table VIII compares it with recent papers.

With 440 test samples, one sample is 0.23 percentage points, so the 0.1–0.4 point differences in Table VIII are not meaningful. The honest reading is that AgriCopilot is on par with the best recent systems on accuracy and that its added value is the probability distribution, SHAP explanation and DS risk context.

C. Soil Fertility

On the 176-sample test split accuracy is 100%, but this is inflated: the engineered threshold features (`N_score`, `P_score`) encode the labelling rule. We therefore report the five-fold cross-validated accuracy of **95.84%** (precision ≈96.1%, recall ≈95.8%, F1 ≈95.9%) as the operational figure.

TABLE IX
YIELD BENCHMARK (T/HA, 300 TEST SAMPLES)

Model	Split	MAE	RMSE	R^2
XGBoost	Train	0.053	0.078	1.000
XGBoost	Val	2.172	3.648	0.9671
XGBoost	**Test**	**2.330**	**3.956**	**0.9632**
Random forest	Test	2.573	4.509	0.9522
OLS	Test	8.774	10.754	0.7284

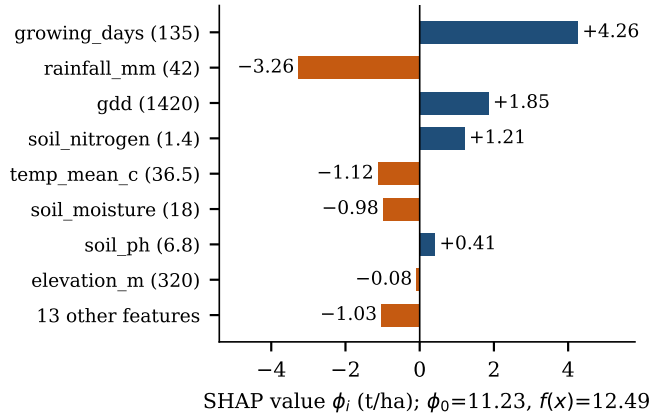


Fig. 2. TreeSHAP attribution for one example field. Predicted yield 12.49 t/ha (90% conformal interval 5.06–17.72 t/ha).

D. Yield Prediction

Table IX shows XGBoost as champion with test $R^2 = 0.9632$ (MAE 2.33, RMSE 3.96 t/ha), +0.235 absolute R^2 over OLS and 63.2% lower RMSE; the random forest follows at 0.9522. The training R^2 of 1.000 indicates over-fitting, which the validation and test scores (0.9671, 0.9632) show to be modest.

Comparison. We deliberately do not rank our R^2 against published yield studies. Our data are synthetic, whereas Li *et al.* [16] work with real multi-year county records and report improvements relative to a baseline, and the rice-yield study of Jagan Mohan *et al.* [17] contains internally inconsistent metrics. What can be compared fairly is capability: both recent studies stop at point predictions (and SHAP/LIME in [17]), whereas AgriCopilot adds a 90% conformal interval. Real-data validation is the first item of future work.

Fig. 2 shows the attribution: a long 135-day season (+4.26) and good thermal accumulation (+1.85) raise the prediction, while a 42 mm rainfall deficit (−3.26) and 36.5 °C mean temperature (−1.12) lower it. The eight listed features sum to +2.29, the remaining 13 contribute −1.03, and $11.23 + 1.26 = 12.49$ t/ha.

E. Multimodal Ablation

Table X and Fig. 3 add modalities one at a time to the unified risk index. Error falls monotonically from 0.4820 to 0.1712 (a 64.5% reduction), Spearman ρ rises from 0.612 to 0.891 and risk-category agreement from 64.2% to 92.4%. Leaf symptoms alone are a weak risk indicator; environmental context supplies most of the improvement.

TABLE X
MULTIMODAL ABLATION OF THE RISK INDEX

Active modalities	MAE	Spearman ρ	Agree (%)
A: Vision	0.4820	0.612	64.2
B: + Weather	0.3210	0.748	78.5
C: + Soil	0.2190	0.835	87.1
D: + NDVI (full)	**0.1712**	**0.891**	**92.4**

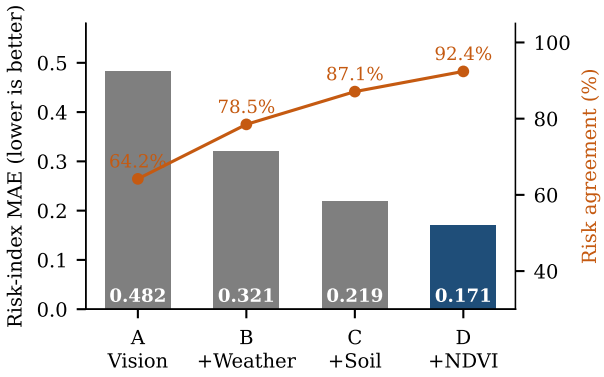


Fig. 3. Ablation: risk-index MAE (bars) and risk-category agreement (line) as modalities are added.

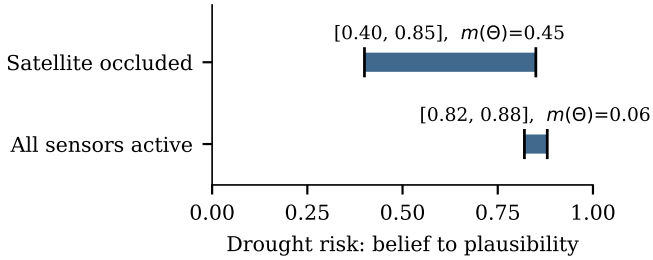


Fig. 4. Belief–plausibility intervals for drought risk with and without usable satellite data.

F. Dempster–Shafer Fusion

Fig. 4 shows two drought-risk outputs. With all sensors active the interval is narrow ($[0.82, 0.88]$, $m(\Theta) = 0.06$). When cloud cover exceeds 50% the satellite source supplies little evidence, and the interval widens to $[0.40, 0.85]$ ($m(\Theta) = 0.45$), warning the user to verify on the ground. For a fully unavailable satellite feed, that source's own assignment moves to $m(\Theta) = 0.80$ instead of a fabricated NDVI value.

G. RAG Copilot

On the project's internal question set the copilot achieved a top-3 retrieval hit rate of 96.2%, MMR diversity 0.84, query latency 180 ms and BLEU-4 of 0.68. AgriRegion [26] reports different metrics (RAGAS-based precision 0.86, on its own 160-question set), so these figures cannot be compared directly;

the relevant difference is that our assistant is conditioned on live field state and sits beside the explainable models.

H. Master Evaluation Matrix

TABLE XI
SUMMARY OF ALL COMPONENTS

Component	Architecture	Result	XAI
Disease 1A	EffNetV2-S, 17 cl.	94.82%, F1 0.945	Grad-CAM
Disease 1B	EffNetV2-S, 38 cl.	99.49%, F1 0.995	Grad-CAM
Crop rec.	RF + XGB vote	99.55%, F1 0.9955	feat. importance
Soil	ET + RF + XGB	95.84% (CV)	nutrient profile
Yield	XGBoost	R^2 0.9632, MAE 2.33	TreeSHAP + conformal
Risk fusion	Dempster–Shafer	92.4% agreement	Bel/Pl intervals
RAG	ChromaDB + LLM	96.2% top-3 hit	citations

X. LIMITATIONS

- The yield model was trained and tested on a *synthetic* 2,000-sample set. Its R^2 shows that the pipeline works, not that it will generalise to real farms.
- The 100% soil test accuracy reflects label leakage through engineered threshold features; only the 95.84% cross-validated figure should be quoted.
- The ablation and RAG numbers come from the authors' internal evaluation sets and have not been independently replicated.
- The crop-recommendation dataset is near-saturated; accuracy differences of a few samples are not significant.
- Comparisons with published work mix different datasets, class counts and protocols, and are indicative only.
- Dempster's rule normalises away conflict; under strongly contradictory evidence the interval should be read together with the reported conflict K.

XI. CONCLUSION AND FUTURE WORK

AgriCopilot shows that disease diagnosis, crop recommendation, soil grading, yield prediction, risk assessment and a conversational advisor can share one explainable pipeline. Its accuracy is on par with the best recent single-task systems (99.49% disease, 99.55% crop recommendation, $R^2 = 0.9632$ yield on synthetic data), and it adds what they lack: three levels of explanation, conformal and evidence-theoretic uncertainty, and graceful behaviour when data are missing. The ablation shows a 64.5% reduction in risk error when weather, soil and NDVI accompany vision. Future work: validate yield on real district records, quantise the CNNs for on-device use, add drone multispectral input, extend the RAG corpus to Tamil and Hindi, and add pest life-cycle models driven by weather time series.

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